

This assignment will not be graded and is only for practice.

Multiple Select Question (MSQ)

1) Consider a function $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ defined as:

1 point

$$f(x, y) = \begin{cases} \frac{2xy}{\sqrt{2(x^2+y^2)}} & \text{if } (x, y) \neq (0, 0) \\ 1 & \text{if } (x, y) = (0, 0). \end{cases}$$

Choose the set of correct options.

- ☐ $|x| \leq \sqrt{x^2 + y^2}$ and $|y| \leq \sqrt{x^2 + y^2}$.
- ☐ If $(x, y) \rightarrow (0, 0)$, then $f(x, y) \rightarrow 0$.
- ☐ $\lim_{(x, y) \rightarrow (0, 0)} f(x, y)$ exists.
- ☐ $\lim_{(x, y) \rightarrow (0, 0)} f(x, y) = 0$.
- ☐ $f(x, y)$ is continuous at the origin $(0, 0)$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$$|x| \leq \sqrt{x^2 + y^2} \text{ and } |y| \leq \sqrt{x^2 + y^2}.$$

If $(x, y) \rightarrow (0, 0)$, then $f(x, y) \rightarrow 0$.

$\lim_{(x, y) \rightarrow (0, 0)} f(x, y)$ exists.

$\lim_{(x, y) \rightarrow (0, 0)} f(x, y) = 0$.

2) Consider the following functions

1 point

$$f(x, y) = (\sqrt{1-x}, \sqrt{y})$$

$$g(x, y) = \sin((x^2 - 1)^2 + y^4)$$

$$h(x, y) = \frac{x}{x^2 + y^2}$$

$$s(x, y) = x^2 e^{y^2}$$

Which of the following is true?

- ☐ Domain of $f = \{(x, y) \mid x \leq 1, y \geq 0\}$
- ☐ Domain of $h = \mathbb{R}^2$
- ☐ $g \circ f(x, y) = \sin(x^2 - 1 + y^2)$
- ☐ $((g \circ f) \times h)(x, y) = \frac{\sin(x^2 + y^2)}{x^2 + y^2}$
- ☐ $\lim_{(x, y) \rightarrow (0, 0)} ((g \circ f) \times h + s)(x, y) = 0$

No, the answer is incorrect.

Score: 0

Accepted Answers:

Domain of $f = \{(x, y) \mid x \leq 1, y \geq 0\}$
 $\lim_{(x, y) \rightarrow (0, 0)} ((g \circ f) \times h + s)(x, y) = 0$

3) Match the functions of two variables in Column A with their graphs given in Column B. **1 point**

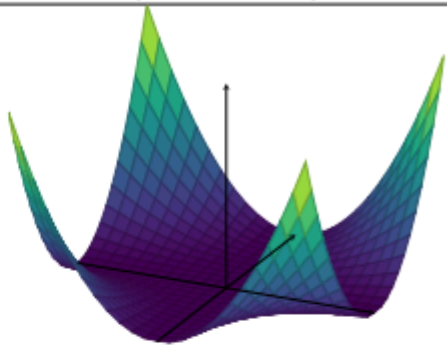
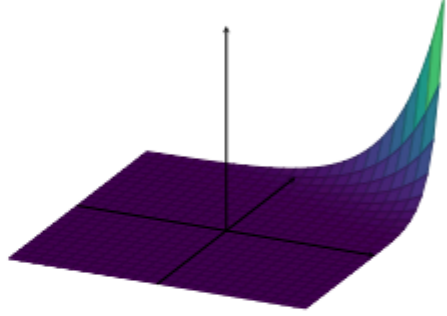
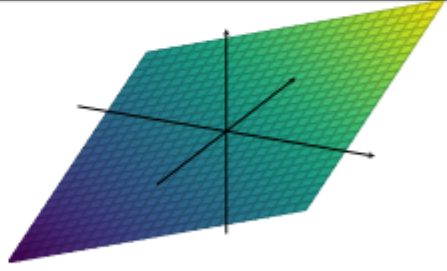
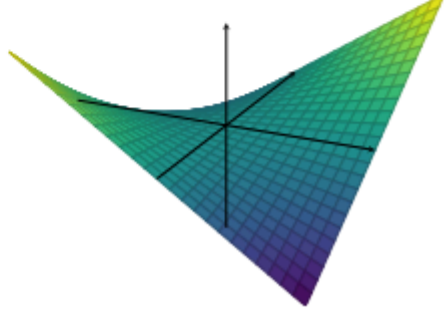
	Function of two variables (Column A)		Graph of the function (Column B)
i)	$f(x, y) = x + y$	1)	
ii)	$f(x, y) = xy$	2)	
iii)	$f(x, y) = x^2 y^2$	3)	
iv)	$f(x, y) = 5^{2x+4y}$	4)	

Table: M2W9P1

- ☐ i $\rightarrow$ 4, ii $\rightarrow$ 3
- ☐ i $\rightarrow$ 3, ii $\rightarrow$ 4
- ☐ iii $\rightarrow$ 1, iv $\rightarrow$ 2
- ☐ iii $\rightarrow$ 2, iv $\rightarrow$ 1

No, the answer is incorrect.

Score: 0

Accepted Answers:

i $\rightarrow$ 3, ii $\rightarrow$ 4

iii $\rightarrow$ 1, iv $\rightarrow$ 2

4) Consider two functions $f : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ and $g : \mathbb{R}^2 \rightarrow \mathbb{R}$ defined as:

1 point

$$f(x, y) = (x^2, y^2)$$

and

$$g(x, y) = \begin{cases} \frac{x^2 - y^2}{x^2 + y^2} & \text{if } (x, y) \neq (0, 0) \\ 0 & \text{if } (x, y) = (0, 0) \end{cases}$$

Which of the following option(s) is (are) true?

- ☐ $g \circ f(x, y) = \begin{cases} \frac{x^2 - y^2}{x^2 + y^2} & \text{if } (x, y) \neq (0, 0) \\ 0 & \text{if } (x, y) = (0, 0) \end{cases}$
- ☐ $g \circ f(x, y) = \begin{cases} \frac{x^4 - y^4}{x^4 + y^4} & \text{if } (x, y) \neq (0, 0) \\ 0 & \text{if } (x, y) = (0, 0) \end{cases}$
- ☐ $\lim_{(x, y) \rightarrow (0, 0)} g(x, y)$ exists.
- ☐ $\lim_{(x, y) \rightarrow (0, 0)} g \circ f(x, y)$ exists.
- ☐ $\lim_{x \rightarrow 0} \lim_{y \rightarrow 0} g \circ f(x, y) = \lim_{y \rightarrow 0} \lim_{x \rightarrow 0} g \circ f(x, y)$
- ☐ $\lim_{x \rightarrow 0} \lim_{y \rightarrow 0} g(x, y) = 1$
- ☐ $\lim_{y \rightarrow 0} \lim_{x \rightarrow 0} g \circ f(x, y) = -1$
- ☐ $\lim_{(x, y) \rightarrow (1, 1)} g(x, y)$ does not exist.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$$g \circ f(x, y) = \begin{cases} \frac{x^4 - y^4}{x^4 + y^4} & \text{if } (x, y) \neq (0, 0) \\ 0 & \text{if } (x, y) = (0, 0) \end{cases}$$

$$\lim_{x \rightarrow 0} \lim_{y \rightarrow 0} g(x, y) = 1$$

$$\lim_{y \rightarrow 0} \lim_{x \rightarrow 0} g \circ f(x, y) = -1$$

Numerical Answer Type (NAT)

5) Consider a function $f(x, y) = xy$. If the directional derivative of f at point (a, b) , in direction of $(1, -1)$ and $(1, 1)$ are 1 and 5 respectively. Then find the value $\frac{2a+b}{\sqrt{2}}$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 7

1 point

6) Consider the following statements:

Statement 1: Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ be a function such that $\lim_{x \rightarrow a} \lim_{y \rightarrow b} f$ exists, where $a, b \in \mathbb{R}$. Then $\lim_{(x,y) \rightarrow (a,b)} f$ exists.

Statement 2: Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ be a function such that $\lim_{y \rightarrow b} \lim_{x \rightarrow a} f = f(a, b)$, where $a, b \in \mathbb{R}$. Then f is continuous.

Statement 3: Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ be a function such that $\lim_{x \rightarrow a} \lim_{y \rightarrow b} f = \lim_{y \rightarrow b} \lim_{x \rightarrow a} f = f(a, b)$, where $a, b \in \mathbb{R}$. Then f is continuous.

Statement 4: Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ be a function such that $f_x(a, b)$ exists. Then f is continuous.

Statement 5: Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ be a function such that $f_y(a, b)$ exists. Then f is continuous.

How many statements are correct?

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: String) 0

1 point

7) Consider the following two functions $f_1, f_2 : \mathbb{R}^2 \rightarrow \mathbb{R}$ defined as:

$$\begin{aligned} f_1(x, y) &= xy + 2x \\ f_2(x, y) &= x^2 + 2y + 3x^2y. \end{aligned}$$

Define a matrix A as follows:

$$A = \begin{bmatrix} \frac{\partial f_1}{\partial x}(3, 0) & \frac{\partial f_2}{\partial x}(1, 1) \\ \frac{\partial f_1}{\partial y}(3, 2) & \frac{\partial f_2}{\partial y}(\sqrt{\frac{7}{3}}, 2) \end{bmatrix}.$$

What will be the value of $\det(A)$?

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) -6

1 point

Comprehension Type Question

An earthworm is crawling about on a sunny day. We draw coordinates axes on the ground where it is crawling and find that the body temperature of the earthworm at a point (x, y) is given by the function:

$$T(x, y) = 2x^2 + 3xy + y^2$$

Use this information to answer questions 8,9, and 10.

8) Which of the following option(s) is (are) true?

1 point

- ☐ $\lim_{(x,y) \rightarrow (1,1)} T(x, y) = \lim_{x \rightarrow 1} \lim_{y \rightarrow 1} T(x, y)$
- ☐ $T(x, y)$ is continuous at $(1, 1)$
- ☐ $T(x, y)$ is a linear function.
- ☐ $T(cx, cy) = c T(x, y)$.
- ☐ If the earthworm approaches the point $(1, 2)$ from any direction, then the body temperature of the earthworm approaches 12 units.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$$\lim_{(x,y) \rightarrow (1,1)} T(x, y) = \lim_{x \rightarrow 1} \lim_{y \rightarrow 1} T(x, y)$$

$T(x, y)$ is continuous at $(1, 1)$

If the earthworm approaches the point $(1, 2)$ from any direction, then the body temperature of the earthworm approaches 12 units.

9) If the gradient of $T(x, y)$ is $(25, 18)$ at a point, then find the body temperature of the earthworm at that point.

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 77

1 point

10) Which of the following option(s) is (are) true?

1 point

- ☐ If the earthworm moves parallel to the X -axis, then rate of change of its body temperature at the point (a, b) is given by $4a + 3b$.
- ☐ If the earthworm moves parallel to the Y -axis, then rate of change of its body temperature at the point (a, b) is given by $2a + 3b$.
- ☐ If the earthworm moves parallel to the straight line joining the origin and the point $(2, 3)$, then rate of change of its body temperature at the point $(1, 1)$ is given by $\frac{29}{\sqrt{13}}$.
- ☐ If the earthworm moves parallel to the straight line joining the origin and the point $(0, 1)$, then rate of change of its body temperature at the point $(1, 2)$ is given by 10.

No, the answer is incorrect.

Score: 0

Accepted Answers:

If the earthworm moves parallel to the X -axis, then rate of change of its body temperature at the point (a, b) is given by $4a + 3b$.

If the earthworm moves parallel to the straight line joining the origin and the point $(2, 3)$, then rate of change of its body temperature at the point $(1, 1)$ is given by $\frac{29}{\sqrt{13}}$.

Your score is: 0/10